WORKED SOLUTION

# STEP III 2018, Question 12

Probability and Statistics • Order Statistics

**STEP III 2018, Question 12**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q12

## The key idea

The event "the $k$th smallest number is at most $y$" is the same as "at least $k$ of the $n$ numbers are at most $y$", and the number of values at most $y$ is binomial. That gives the distribution function straight away. Differentiating it produces two sums, and the two binomial identities in (ii) are exactly what makes them telescope. Once we know the density, the fact that it integrates to 1 evaluates an integral for free, and the same integral (with $n$ and $k$ shifted) gives the mean.

## Solution

Call the $n$ numbers $X_1, \ldots, X_n$. They are independent, each uniform on $[0,1]$.

### Part (i)

Fix $y$ with $0 \le y \le 1$ and let $N$ be the number of the $X_i$ with $X_i \le y$. Each $X_i$ satisfies $X_i \le y$ with probability $y$, independently of the others, so $N \sim \mathrm{B}(n, y)$.

Now $Y_k \le y$ exactly when at least $k$ of the numbers are at most $y$. If $Y_k \le y$, then $Y_k$ and the $k-1$ numbers below it are all at most $y$, so $N \ge k$. Conversely, if $N \ge k$, then the $k$ smallest numbers are all at most $y$, and in particular $Y_k \le y$. Hence

$$\mathrm{P}(Y_k \le y) = \mathrm{P}(N \ge k) = \sum_{m=k}^{n} \binom{n}{m} y^m (1-y)^{n-m},$$

which is $(*)$.

### Part (ii)

Writing out the factorials, for $1 \le m \le n$,

$$m\binom{n}{m} = \frac{n!}{(m-1)!\,(n-m)!} = n \cdot \frac{(n-1)!}{(m-1)!\,((n-1)-(m-1))!} = n\binom{n-1}{m-1},$$

and similarly, for $0 \le m \le n-1$,

$$(n-m)\binom{n}{m} = \frac{n!}{m!\,(n-m-1)!} = n \cdot \frac{(n-1)!}{m!\,((n-1)-m)!} = n\binom{n-1}{m}.$$

(When $m = n$ the left-hand side is 0, and so is the right if we read $\binom{n-1}{n}$ as 0.)

**The density.** The density $f(y)$ of $Y_k$ is the derivative of $(*)$. Differentiating each term by the product rule,

$$f(y) = \sum_{m=k}^{n} \binom{n}{m} \Big[ m\, y^{m-1}(1-y)^{n-m} - (n-m)\, y^m (1-y)^{n-m-1} \Big].$$

In the second piece the $m = n$ term vanishes because of the factor $n - m$, so that sum really runs to $m = n-1$. Using the two identities,

$$f(y) = n \sum_{m=k}^{n} \binom{n-1}{m-1} y^{m-1}(1-y)^{n-m} \; - \; n \sum_{m=k}^{n-1} \binom{n-1}{m} y^m (1-y)^{n-1-m}.$$

**Ask yourself** These two sums look almost the same. How can we make them line up?

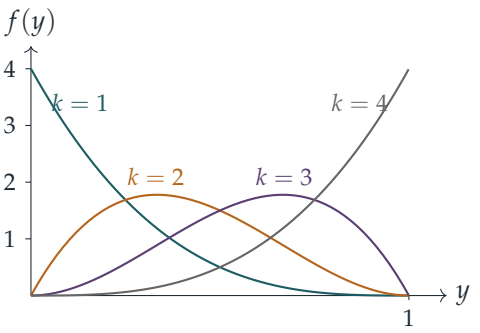
Put $j = m+1$ in the second sum. It becomes

$$n \sum_{j=k+1}^{n} \binom{n-1}{j-1} y^{j-1}(1-y)^{n-j},$$

which is the first sum term for term, except that it starts at $k+1$ rather than $k$. Everything cancels except the $m = k$ term of the first sum:

$$f(y) = n\binom{n-1}{k-1} y^{k-1}(1-y)^{n-k},$$

as required. (If $k = n$ the second sum is empty and the first has only the term $m = n$, so the result still holds.)




The densities for $n = 4$: the $k$th smallest of four numbers piles up near $k/5$.

**The integral.** A density integrates to 1 over its range, which here is $[0,1]$. Therefore

$$n\binom{n-1}{k-1}\int_0^1 y^{k-1}(1-y)^{n-k}\,\mathrm{d}y = 1,$$

so

$$\int_0^1 y^{k-1}(1-y)^{n-k}\,\mathrm{d}y = \frac{1}{n\binom{n-1}{k-1}} = \frac{(k-1)!\,(n-k)!}{n!}.$$

This holds for all integers $1 \le k \le n$.

#### Part (iii)

By definition,

$$\mathrm{E}(Y_k) = \int_0^1 y\,f(y)\,\mathrm{d}y = n\binom{n-1}{k-1}\int_0^1 y^k(1-y)^{n-k}\,\mathrm{d}y.$$

> **Ask yourself** This integral isn't quite the one we just found. Which values of $n$ and $k$ turn it into that form?

The integrand is $y^{K-1}(1-y)^{N-K}$ with $K = k+1$ and $N = n+1$, and $1 \le K \le N$, so the formula from (ii) applies:

$$\int_0^1 y^k(1-y)^{n-k}\,\mathrm{d}y = \frac{k!\,(n-k)!}{(n+1)!}.$$

Since $n\binom{n-1}{k-1} = \dfrac{n!}{(k-1)!\,(n-k)!}$,

$$\mathrm{E}(Y_k) = \frac{n!}{(k-1)!\,(n-k)!}\cdot\frac{k!\,(n-k)!}{(n+1)!} = \frac{k}{n+1}.$$

## Checks

- With $n=1$, $k=1$ we have a single uniform number, and $\mathrm{E}(Y_1) = \frac{1}{2}$, as it should be.
- Symmetry: replacing every $X_i$ by $1-X_i$ turns the $k$th smallest into the $k$th largest, so $\mathrm{E}(Y_k) + \mathrm{E}(Y_{n+1-k}) = 1$. Indeed $\frac{k}{n+1} + \frac{n+1-k}{n+1} = 1$.
- The $n$ numbers cut $[0,1]$ into $n+1$ gaps, and the answer says each gap has mean length $\frac{1}{n+1}$, which is very plausible by symmetry.
- With $n=5$, $k=2$: $(*)$ at $y=0.3$ gives 0.47178, and a computer simulation of 200 000 samples gave 0.4726 and a sample mean of $Y_2$ of 0.33333, against $\frac{1}{3}$.

## Common slips

- Writing $\mathrm{P}(Y_k \le y) = \binom{n}{k}y^k(1-y)^{n-k}$, the probability of *exactly* $k$ values below $y$, instead of *at least* $k$.

- Forgetting that the $m = n$ term of the second sum is zero, which leaves the telescoping looking as if it fails by one term.
- Shifting only $k$ (and not $n$) when reusing the integral in (iii).

## Beyond STEP

The $Y_k$ are the *order statistics* of the sample, and $Y_k$ has a Beta distribution with parameters $k$ and $n-k+1$. The integral in (ii) is the Beta function $B(a,b) = \int_0^1 y^{a-1}(1-y)^{b-1}\,\mathrm{d}y = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$, which you will meet again alongside the Gamma function. Order statistics underlie the sample median and non-parametric statistics. Blitzstein and Hwang, *Introduction to Probability*, treats order statistics and the Beta distribution beautifully, including a "story" proof of the Beta integral much like the argument here.

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